



Capstone Project Proposals: Lightning Talk #1

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Data Science bootcamp – General Assembly

Goal: Evaluate 3 candidate datasets to select an impactful Capstone project.

Topic A

Factors Affecting
Children Anemia Level
(Classification / Public
Health)

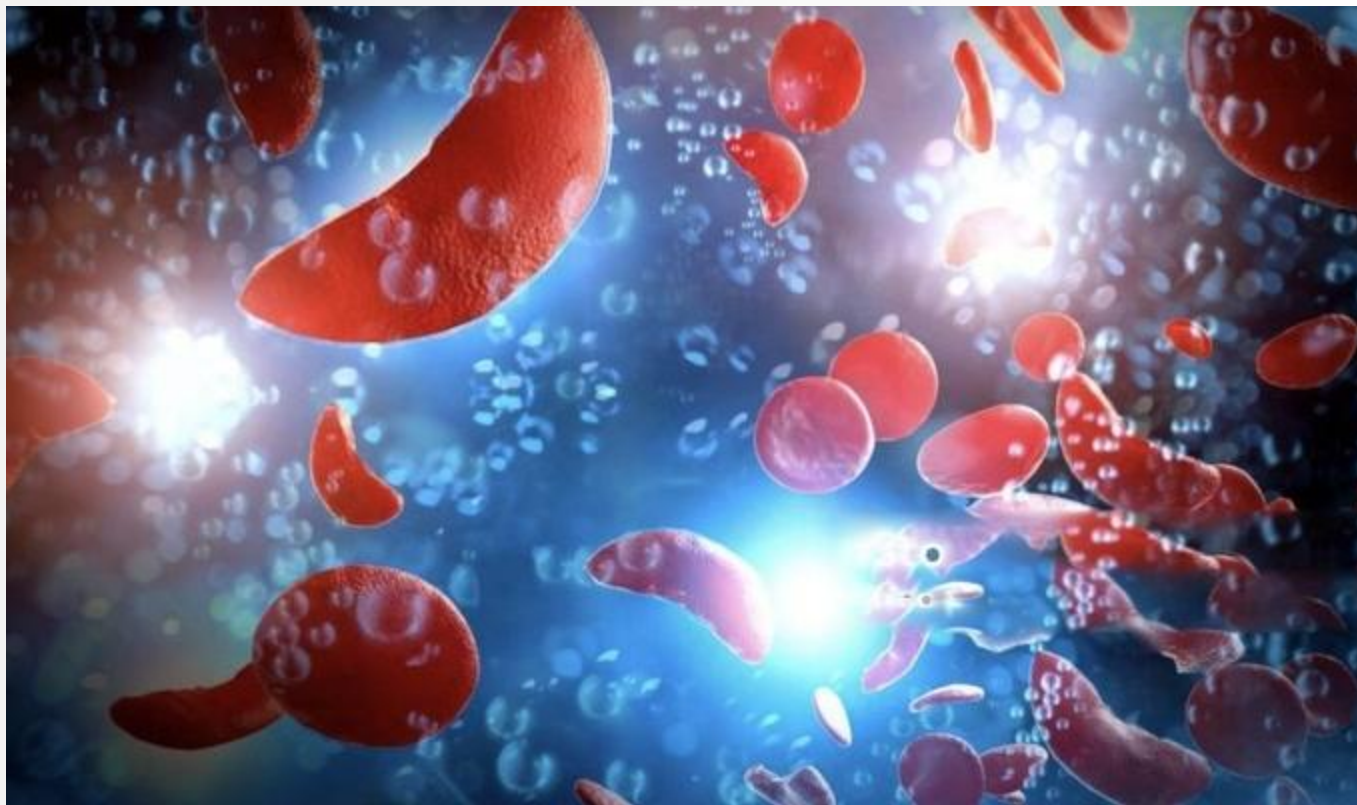
Topic B

FoodMart Media Cost
Prediction (Regression /
Retail)

Topic C

Stellar Classification
SDSS DR17
(Classification /
Astronomy)

Factors Affecting Children Anemia Level



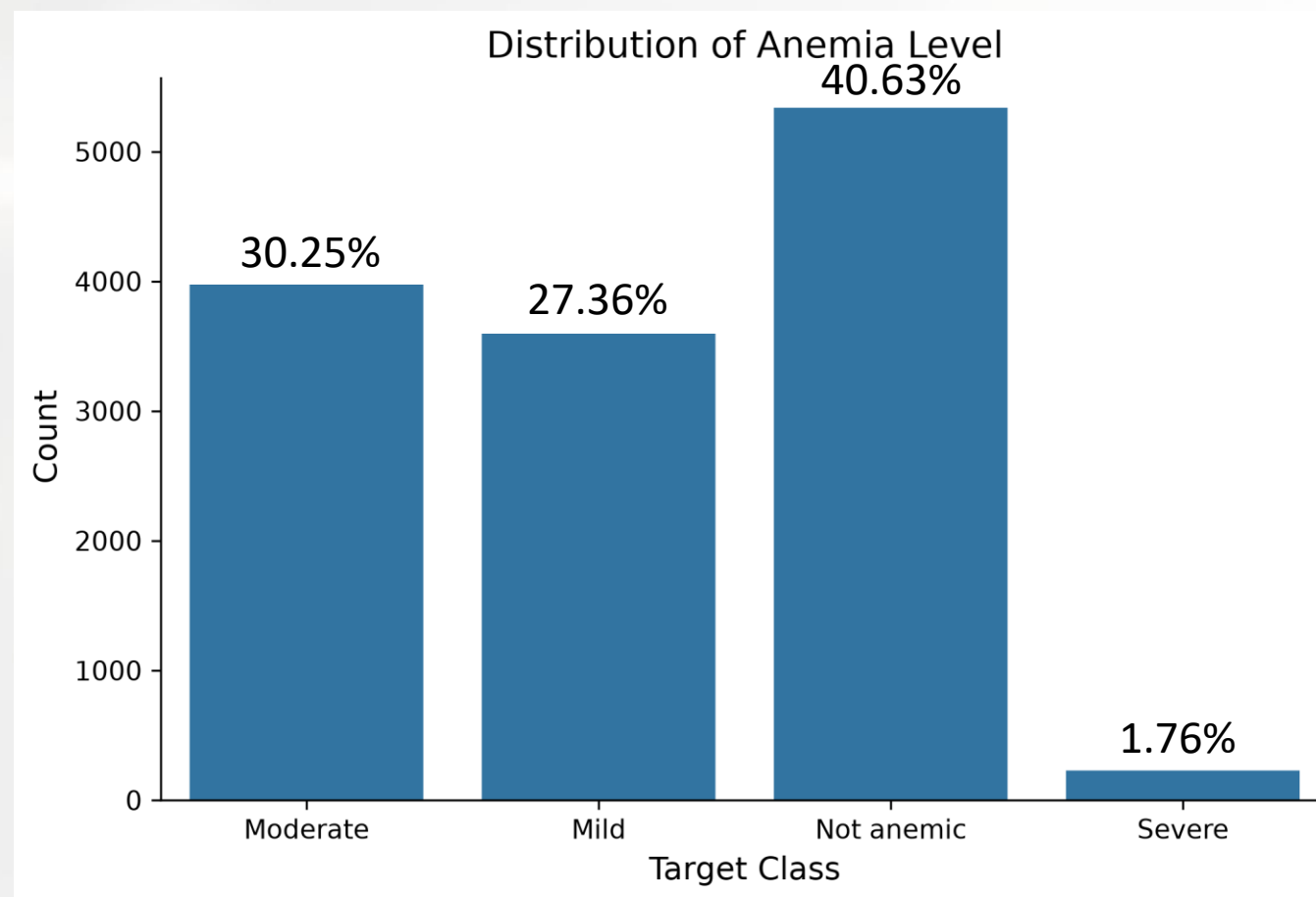
Problem Statement

Childhood anemia is a major public health challenge driven by a complex mix of factors. Identifying key risk drivers is essential for public health organizations to deploy targeted interventions effectively.

Goals and aims

Build a classification model to predict child anemia severity levels and help health organizations optimize resource allocation.

Insights about the data



1

There are missing values in 8 columns, the most notable are Anemia level, and Hemoglobin level adjusted for altitude. Both have many missing values.

2

The number of numerical columns = 4
The number of categorical columns = 12

3

Many columns have outliers such as: Age at 1st birth, Births in last five years, Hemoglobin level adjusted for altitude and smoking

FoodMart Media Cost Prediction



Problem Statement

Retail chains lose substantial capital on inefficient promotional campaigns because marketing cost varies drastically across store layouts, media types, and customer demographics.

Goals and aims

Build a regression model to accurately predict promotional media costs (Cost), allowing marketing managers to optimize advertising budget allocation.

1

There are no missing values in any of the columns of the dataset

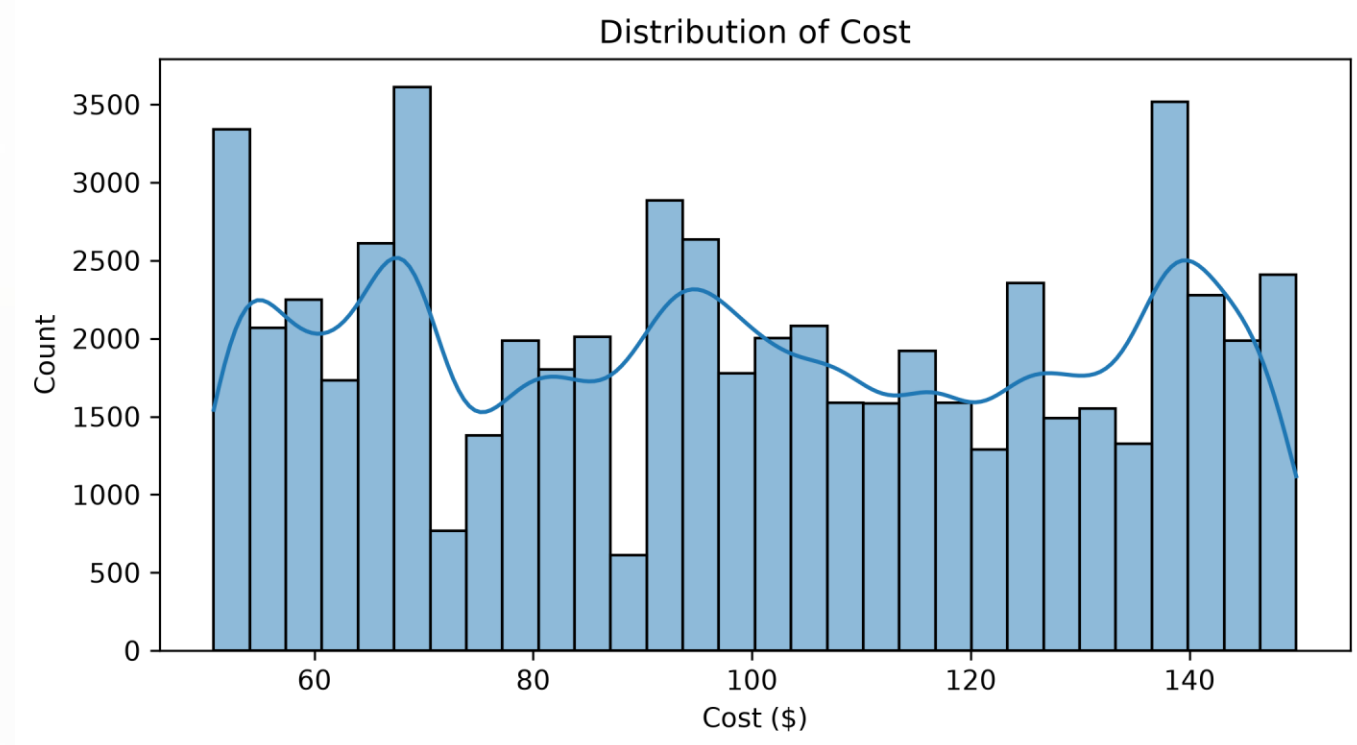
2

The number of numerical columns = 22
The number of categorical columns = 17

3

6 columns have outliers: meat_sqft, frozen_sqft, num_children_at_home, store_cost(in millions), unit_sales(in millions), store_sales(in millions)

Insights about the data



Stellar Classification

Dataset - SDSS17



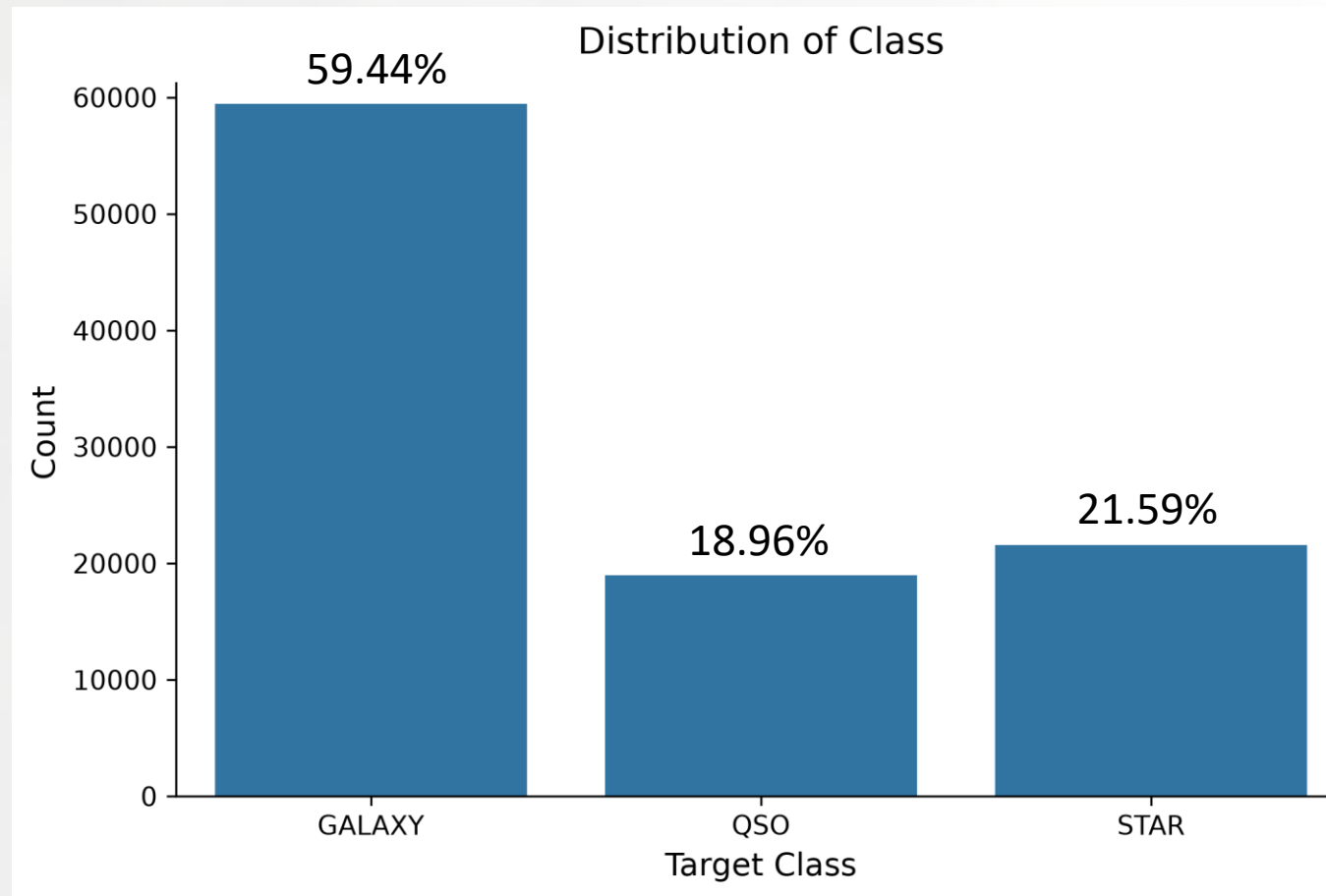
Problem Statement

Modern automated space telescopes collect massive volumes of sky survey data, making manual optical inspection far too slow and labor intensive for astronomical research.

Goals and aims

Build a classification model using photometric spectral filter data and redshift to classify space objects rapidly.

Insights about the data



1

There are no missing values in any of the columns of the dataset

2

The number of numerical columns = 17
The number of categorical columns = 0

3

7 columns have outliers: redshift, field_ID, z (infrared), i (near-infrared), r (red), g (green), u (ultraviolet)

A top-down view of a desk with a laptop, a cup of coffee, a pen, glasses, a paperclip, and a monstera leaf.

THANK YOU